Week 1 — Cybersecurity Lab Setup

Project Module 1 • WK1-PM1

VirtualBox and Kali Linux 2026.3 | Lab configuration and verification

Prepared by	Beamlak Fiseha Wagaw
Lab platform	Oracle VirtualBox on Windows
Guest operating system	Kali Linux 2026.3 (64-bit)
Reporting date	31 October 2026
Scope	Phase 1: Kali virtual machine, NAT Network, file sharing, and snapshot

Introduction

For this Week 1 task, I set up a separate Kali Linux 2026.3 virtual machine in VirtualBox. I kept my university Kali installation unchanged and used the new machine for the NetworkWalks lab.

The screenshots confirm the static IP, successful ping to 8.8.8.8, and saved snapshot. The final output for /downloads and the completed google.com ping was not captured.

1. Lab requirements

Virtualization platform	Oracle VirtualBox
Attacking machine	Kali Linux 2026.3
NAT Network IPv4 prefix	10.0.0.0/24
Kali IPv4 address	10.0.0.2/24
Internet access	External IP ping successful; DNS result not fully captured
Guest integration	Bidirectional clipboard and drag-and-drop configured
Host share	Windows Downloads mapped in Kali; /downloads listing not yet captured
Recovery	Week1-NetworkWalks-Completed snapshot created

2. Virtual machine preparation

I downloaded the pre-built Kali Linux 2026.3 VirtualBox image and extracted the files. I then opened the .vbox file in VirtualBox to add the new machine. I did not use the ISO installer for this setup.

The new machine is named kali-linux-2026.3-virtualbox-amd64. VirtualBox shows 2 CPUs, 2048 MB RAM and an approximately 80 GB virtual disk.

I started the machine and logged in to the Kali desktop. I used the terminal for the network setup.

The screenshots in this report are from the internship lab. I have not included my university account details or any passwords.

3. NAT Network configuration

I created a NAT Network named NetworkWalksNAT in VirtualBox. I changed its IPv4 prefix to 10.0.0.0/24, enabled DHCP and left IPv6 disabled.

I opened the new Kali machine's Network settings, selected NAT Network for Adapter 1, chose NetworkWalksNAT and enabled Virtual Cable Connected.

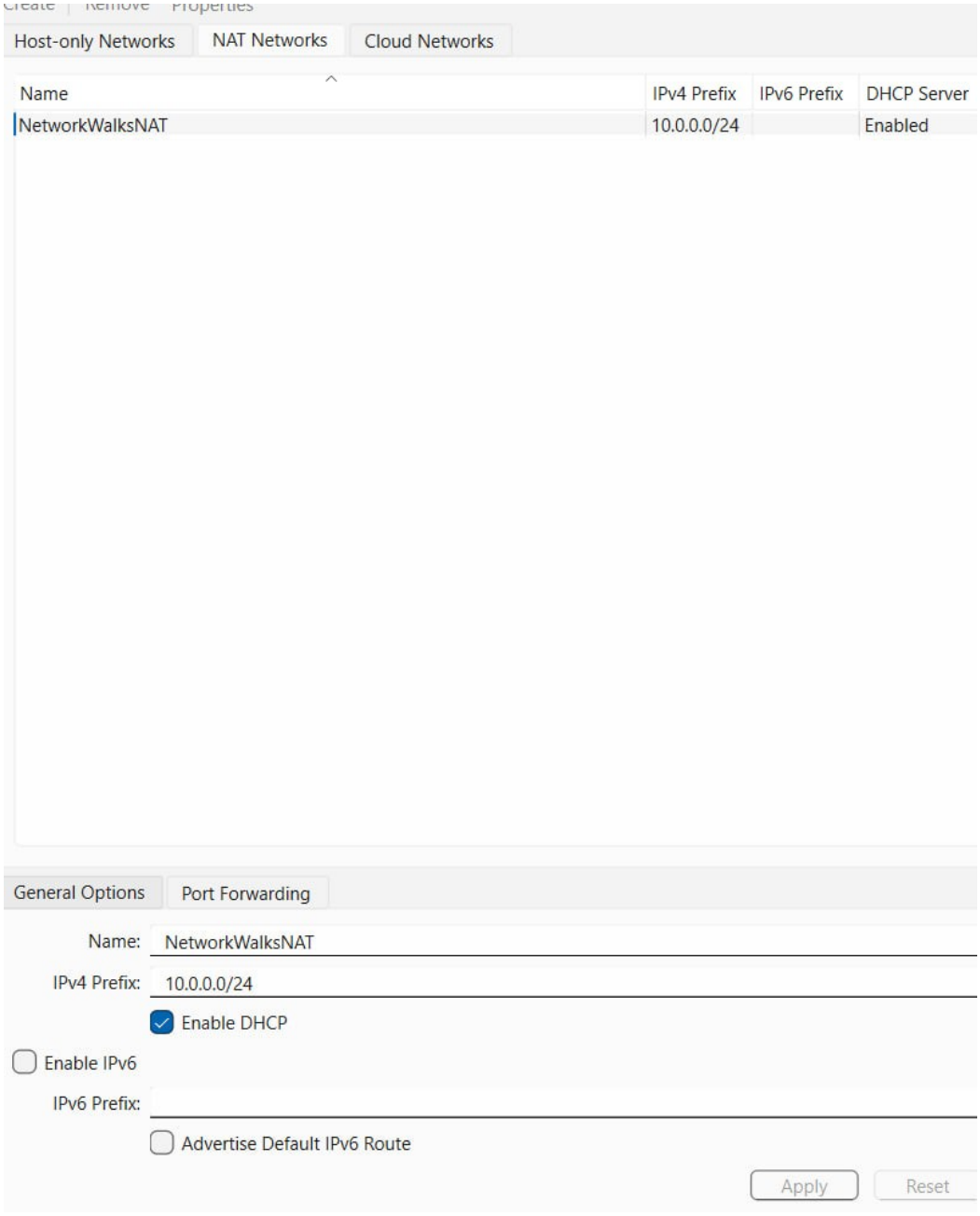


Figure 1. NetworkWalksNAT created with IPv4 prefix 10.0.0.0/24 and DHCP enabled.

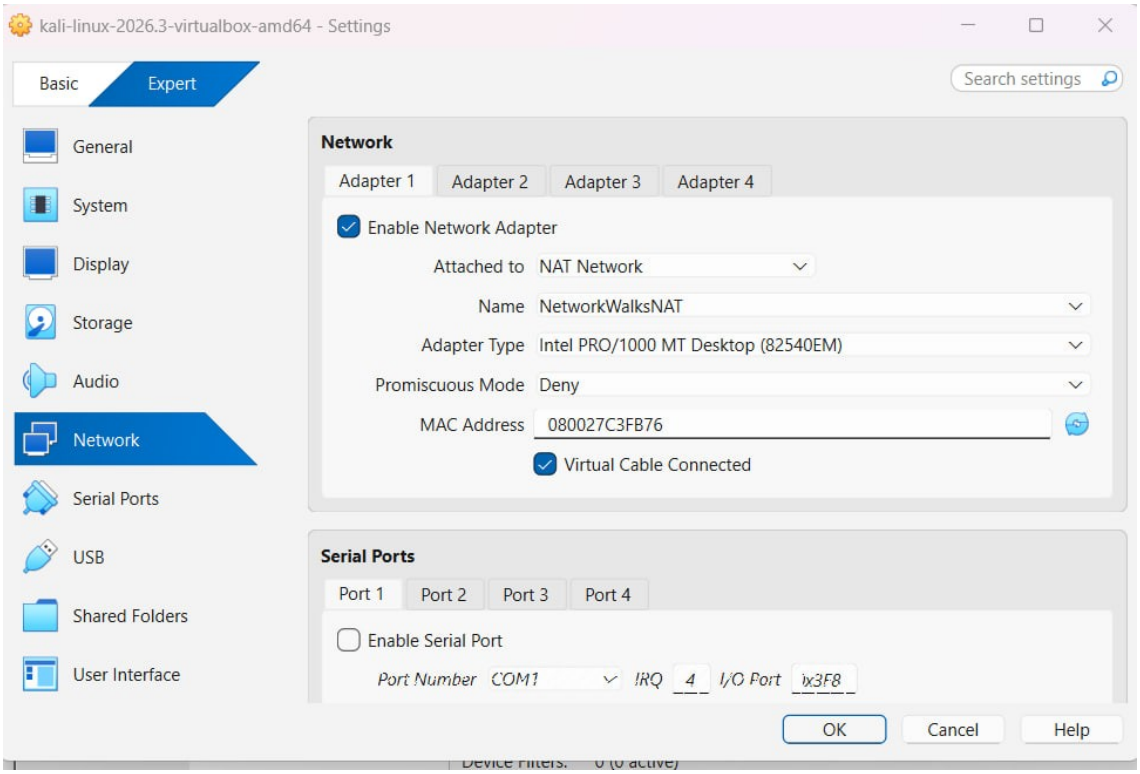


Figure 2. Kali Adapter 1 attached to the custom NAT Network.

4. Static IP assignment and connectivity verification

In Kali, the Ethernet connection was named Wired connection 1 and used eth0. I set its IPv4 address to 10.0.0.2/24, gateway to 10.0.0.1 and DNS to 8.8.8.8 and 1.1.1.1. The connection initially failed. I followed the Week 1 troubleshooting instructions by setting ipv4.dad-timeout to 0, then activated the connection successfully.

Commands used

nmcli connection show

sudo nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.addresses 10.0.0.2/24 ipv4.gateway 10.0.0.1 ipv4.dns "8.8.8.8 1.1.1.1"

sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0

sudo nmcli connection up "Wired connection 1"

ip -4 addr show eth0

ping -c 4 8.8.8.8

ping -c 4 google.com

The terminal shows eth0 assigned 10.0.0.2/24. Four packets were sent to 8.8.8.8 and all four were received (0% packet loss). The google.com ping was started, but its final result is not visible in the screenshot.

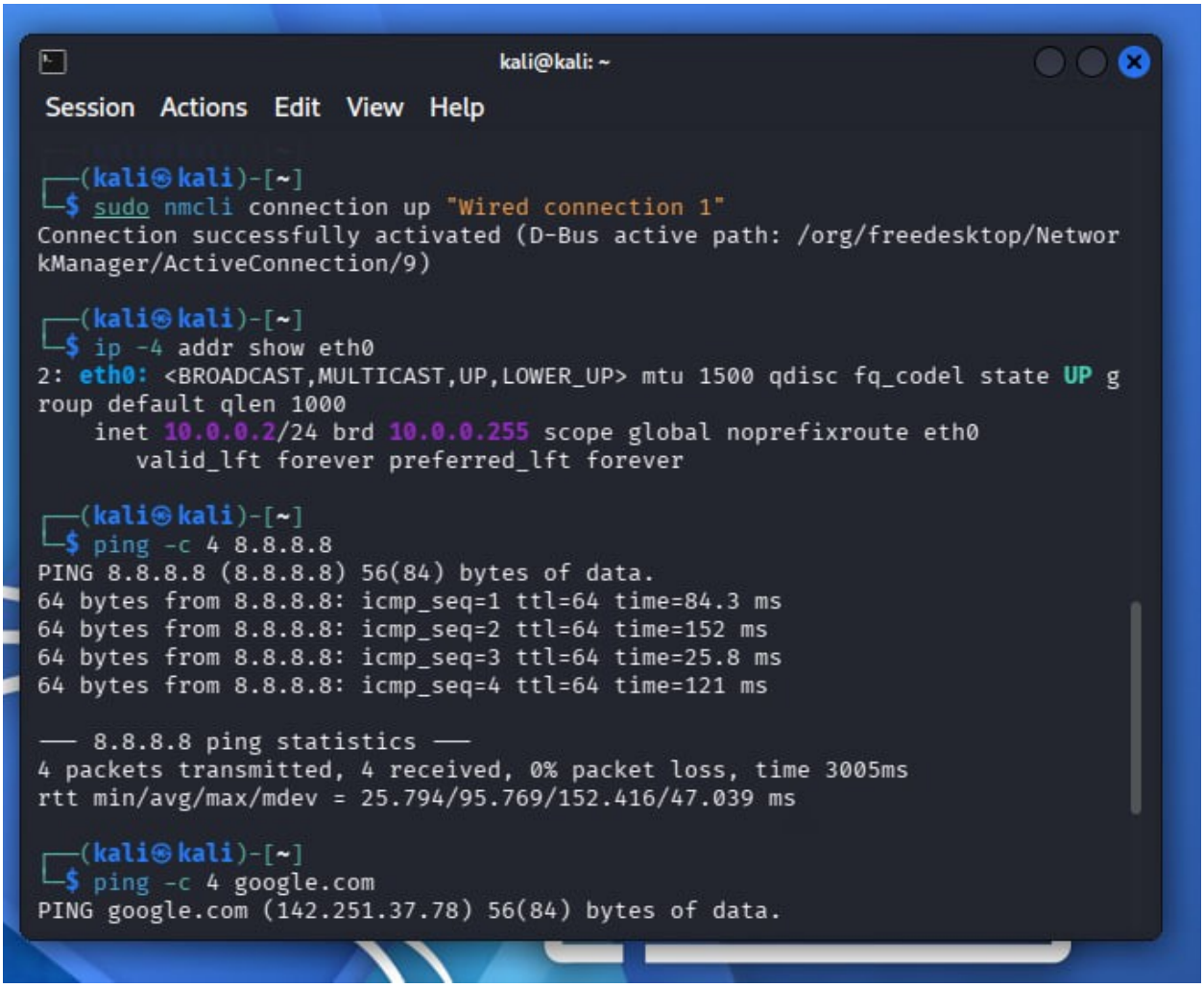


Figure 3. Successful connection activation, static IPv4 address, and external IP ping.

5. Guest integration and shared folder

5.1 Clipboard and drag-and-drop

I set Shared Clipboard and Drag-and-Drop to Bidirectional in VirtualBox Settings > General > Features. The settings are shown in Figure 4.

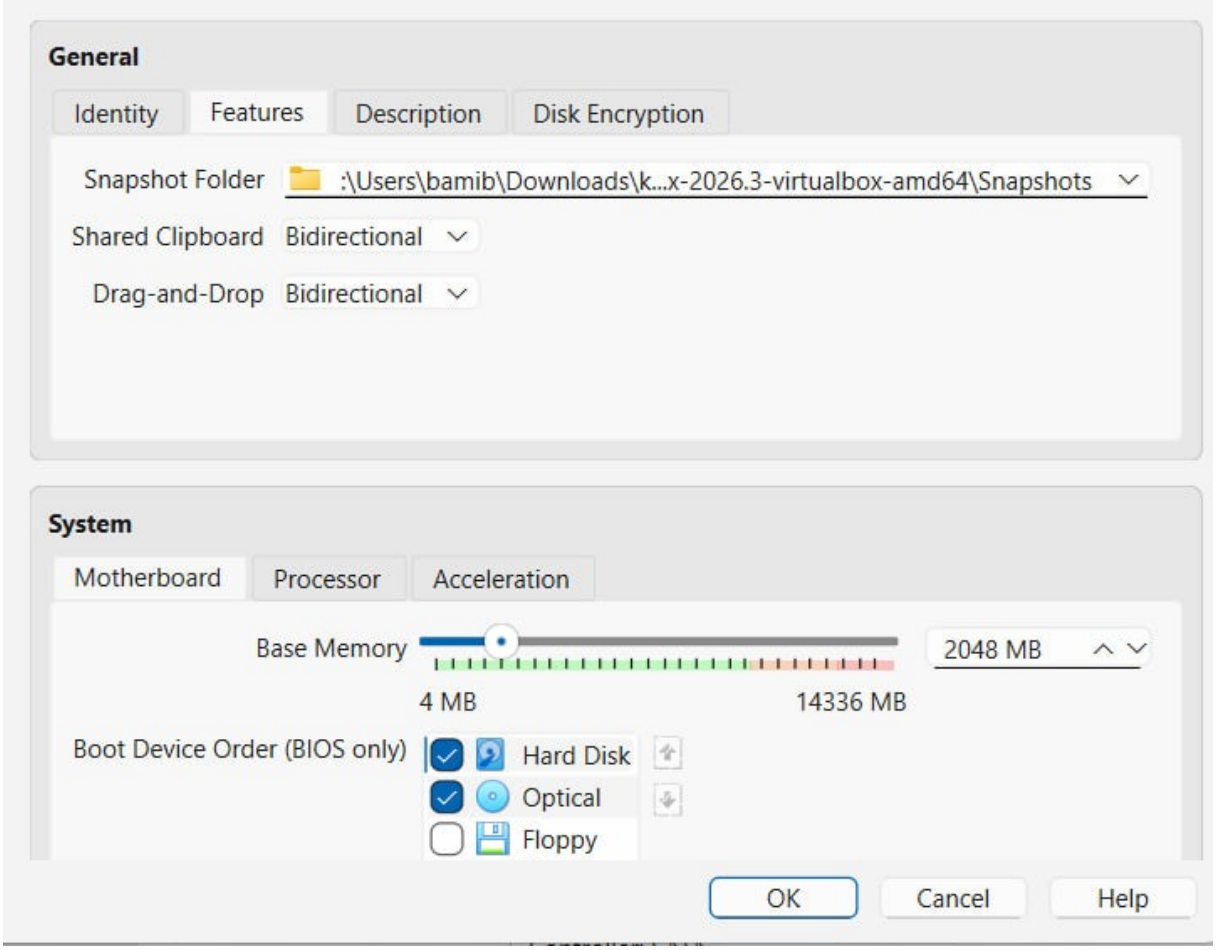


Figure 4. Bidirectional clipboard and drag-and-drop configured.

5.2 Downloads shared folder

I added my Windows Downloads folder as a shared folder with Full access and Auto Mount enabled. In Kali it appeared as sf_Downloads under /media. I then ran a command to link it to /downloads.

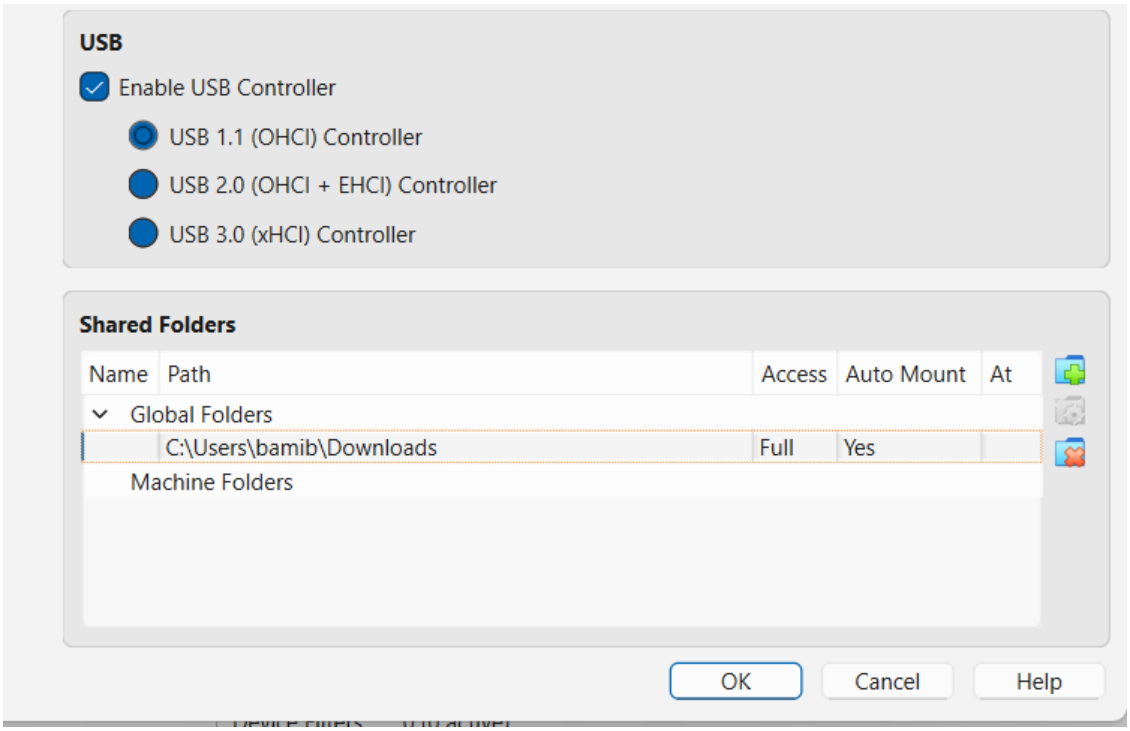


Figure 5. Windows Downloads configured as a VirtualBox Global Folder, Full access, Auto Mount: Yes.

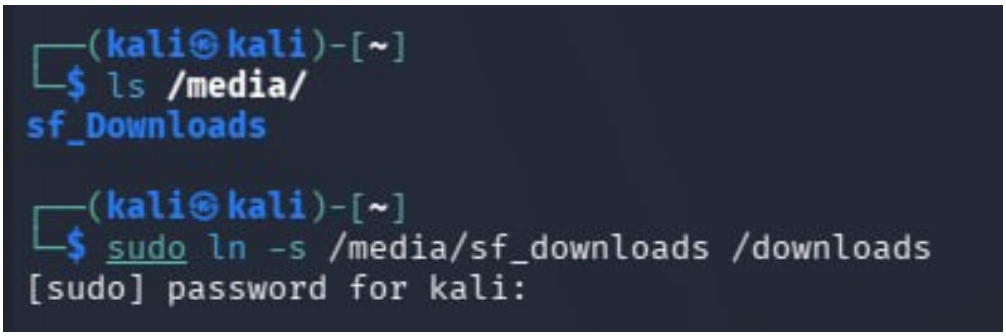


Figure 6. Kali lists sf_Downloads in /media and shows the /downloads link command.

The screenshot shows the shared folder and the link command, but not the final /downloads listing. To verify it, I would run ls -l /downloads and ls /downloads.

6. Recovery snapshot

After the setup, I shut down Kali and saved the snapshot Week1-NetworkWalks-Completed. Figure 7 shows it in the Snapshots tab.

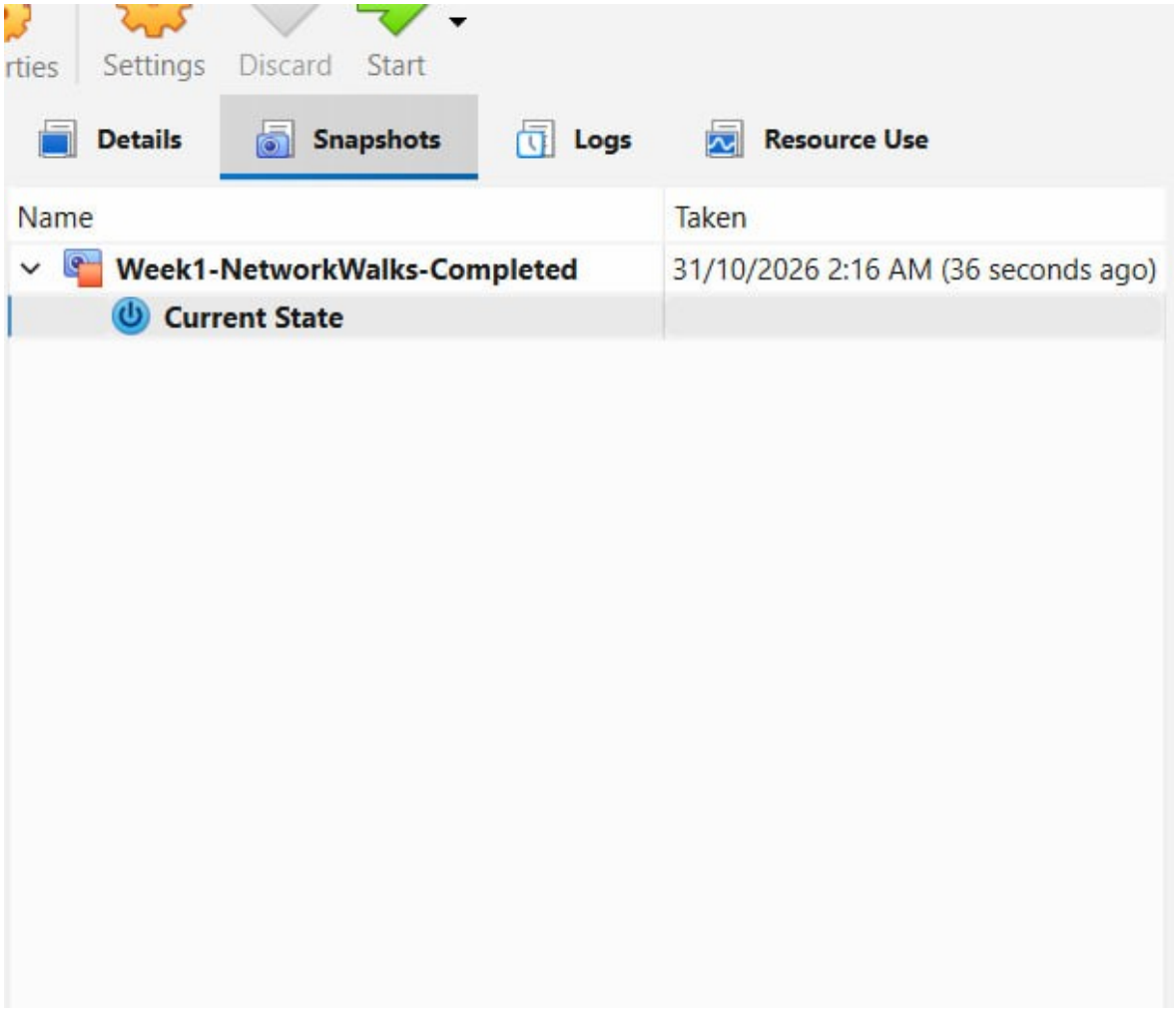


Figure 7. Saved Week1-NetworkWalks-Completed snapshot in VirtualBox.

7. Verification and results

Kali VM imported and running	Confirmed by earlier VirtualBox and Kali desktop evidence
Custom NAT Network 10.0.0.0/24	Confirmed — Figure 1
Adapter connected to NAT Network	Confirmed — Figure 2
Static Kali IP 10.0.0.2/24	Confirmed — Figure 3
External IP connectivity	Confirmed — 4/4 replies from 8.8.8.8, 0% packet loss
DNS/hostname connectivity	Command initiated; completion result not captured
Bidirectional clipboard and drag/drop	Settings confirmed — Figure 4; functional tests not pictured
Shared folder /media/sf_Downloads	Directory visible — Figure 6
Exact /downloads access	Link command shown; successful listing not captured
Snapshot created	Confirmed — Figure 7

8. Conclusion

I completed the main Week 1 lab setup using a separate Kali Linux 2026.3 machine. The screenshots show the NAT Network, static IP, successful external IP ping, sharing settings and saved snapshot. I should still capture the completed google.com ping and the /downloads listing before final submission.

Reference

NetworkWalks Academy. “WK1-PM1 — Week 1 Project Module 1: Lab Setup VirtualBox and Kali Linux,” version 4.2, user-provided task sheet, pp. 1–8.